

Caching in Named Data Networking for the Wireless Internet of Things

Mohamed Ahmed Hail*, Marica Amadeo[†], Antonella Molinaro[†], Stefan Fischer*

*Telematics Institute University of Lübeck, Germany, E-mail: {hail, fischer}@itm.uni-luebeck.de

[†]University Mediterranea of Reggio Calabria, Italy, E-mail: name.surname@unirc.it

Abstract—*Named Data Networking (NDN) is a promising Information-Centric future Internet architecture. Besides its recognized potentialities as a content retrieval solution in wired domains, NDN has been also recently considered as an enabling technology for the Internet of Things (IoT), thanks to its innovative features like named-based routing and in-network caching. In particular, the possibility of caching at intermediate nodes can be especially useful to reduce the retrieval delay, and limit the network traffic and the load on data producer. However, unlike traditional Internet contents, IoT data are typically transient and periodically refreshed by the producer. At the same time, unlike Internet routers, IoT devices can be resource-constrained, with limitations in terms of energy, storage and processing capabilities. Therefore, caching algorithms designed for (intransient) Internet traffic and Internet routers do not well suit IoT domains.*

In this paper, we consider a wireless NDN-IoT network and propose a novel distributed probabilistic caching strategy that relies on the freshness of data and on potentially constrained capabilities of devices (energy level and storage capacity). The proposed solution has been evaluated through simulations with ndnSIM and results show that it outperforms traditional NDN caching mechanisms in terms of data retrieval and network energy efficiency.

Index Terms—Named Data Networking, Information-Centric Networking, Internet of Things, In-network caching, Freshness

I. INTRODUCTION

In the Internet of Things (IoT) vision, heterogeneous wireless devices, ranging from tiny, constrained sensors and actuators to more powerful smartphones and networked vehicles, will interact with the physical surrounding environment and with the Internet to support a wide range of context-aware services. Compared to traditional Internet hosts, like routers, PCs and laptops, most IoT devices are limited in terms of energy, memory, and computational power and they can be also mobile. Moreover, unlike Internet contents characterized by typically large, time-invariant files, IoT data are usually small and transient. New information may be constantly generated, e.g., sensors periodically update environmental data, and applications are interested in the freshest data packets.

Due to such distinct peculiarities of IoT systems, the research community is working on the design of new networking protocols that effectively support IoT communications. In addition to IP-based solutions and cloud-based approaches [1], also the Information-Centric Networking (ICN) paradigm [2] has been recently considered for the IoT. The potentialities of this revolutionary architecture, in fact, are manifold, as observed by the ICN Research Group (ICNRG) of the Internet

Research Task Force (IRTF) [3] and recent literature [4]. In ICN, every content packet is named by using a unique, persistent, location-independent name and is treated by the network nodes as a self-identifying and self-authenticating data unit. The content is retrieved by consumers by directly using its name, without the need of the producer's IP address. This paradigm shift from the IP-based host-centric model to an information-centric one implies several benefits including easier data retrieval and sharing, native mobility support, in-network caching, and content-based security, which are especially useful in IoT. In particular, in-network caching can speed up the data retrieval and reduce the traffic in the network, since data requests could be satisfied by intermediate nodes closer to consumers. This also implies that in-network caching limits the massive data access to the producer and decouples this latter from the receiver(s) by also relaxing the need of continuous connectivity.

Although in-network caching has been widely investigated in ICN wired networks [5], due to the stringent requirements in terms of information *freshness* and the presence of resource-constrained devices, traditional caching algorithms designed for the Internet are not well suited for the IoT. Only a few preliminary works [6]–[8] consider caching in IoT systems but do not propose a specific caching strategy for the wireless environment.

In order to fill this gap, in this paper we focus on the design of a novel caching strategy specifically tailored to wireless multi-hop information-centric IoT systems. Among several ICN instantiations, we refer to the Named Data Networking (NDN) architecture [9], which leverages the simple but robust transmission of *Interest* and *Data* packets used to, respectively, request and deliver named content units. NDN is open to different (new) caching algorithms implementations, i.e., nodes can cache incoming Data packets on the basis of user-defined policies/algorithms.

In this context, we define pCASTING, a *probabilistic Caching Strategy for the Internet of thinGs*, which considers the data freshness and two device features, namely energy level and storage capability, to dynamically adapt the caching probability of each node. The proposed strategy is totally distributed and does not require any additional signalling. Simulation results obtained with ndnSIM [10] show that pCASTING increases the energy efficiency of the network by also guaranteeing lower retrieval delays, when compared with reference caching schemes widely used in NDN, including

universal caching and random caching.

The rest of the paper is organized as follows. Section 2 introduces the NDN architecture and related work about caching in NDN-IoT systems. Section 3 presents the pCASTING proposal, while Section 4 discusses the performance evaluation. Finally, Section 5 concludes the paper.

II. BACKGROUND AND MOTIVATIONS

A. Named Data Networking

NDN has been proposed as a content dissemination architecture for the future Internet that employs hierarchical URI-like content names, carried in Interest and Data packets. Every NDN node maintains three tables: (i) the Content Store (CS), used to cache incoming Data; (ii) the Pending Interest Table (PIT), used to keep track of the forwarded Interest(s) that are not yet satisfied; (iii) the Forwarding Information Base (FIB), used as a routing table based on content names. When the node receives an Interest, it first looks in its CS; if a match is found, then the node sends the Data back. Otherwise, if there is a matching entry in the PIT, the Interest is discarded because an equal request has already been forwarded. Otherwise, a new PIT entry is created and the Interest is further forwarded to the interface stored in the FIB. The Data packet follows the chain of PIT entries back to the consumer(s), and it can be cached in intermediated nodes.

The caching system in a node consists of (i) the caching decision strategy, which specifies whether caching an incoming Data packet; (ii) the replacement policy, which specifies what element must be erased when a new incoming Data packet should be cached and the CS is full. The default caching strategy in NDN is the so called *caching everything everywhere* (CE^2) scheme: nodes cache every incoming Data, while the replacement is carried out by standard policies such as Least-Recently-Used (LRU). However, many papers have already pointed out the inefficiency of the universal caching strategy, e.g., [11], due to the high level of content redundancy and the inefficient utilization of available cache resources. This is why other solutions are usually used in the literature, including probabilistic caching [12]. The key idea behind the probabilistic strategy is that a NDN node randomly caches incoming Data with a certain probability p , where $0 < p < 1$. The CE^2 strategy is a special case of the probabilistic scheme, where $p = 1$. Decreasing p reduces the probability of in-network caching and maximizes the diversity of Data cached in the network.

As discussed in [6], NDN can be very effective in IoT and it can operate above the link layer without requiring an overlay over the IP stack, which would imply additional memory costs. Recent works have started to apply NDN as a clean-slate approach in different IoT scenarios, including Building Management Systems [13] and Intelligent Transportation Systems [14], with focus on naming, security and forwarding aspects.

B. Caching in NDN-IoT systems

A large number of caching strategies have been proposed for NDN networks so far [5], but none of them specifically

addresses IoT systems and their features. They are typically designed for pre-recorded multimedia files like Youtube videos and similar large contents which are frequently requested by users on the Internet; and they usually consider fixed and wiredly connected content routers [11]. It is also worth noticing that the higher performance of such schemes is usually obtained at the expenses of large overhead and complexity [12], which make them unsuitable in presence of resource-constrained (mobile) devices.

When we specifically focus on IoT, the problem of NDN in-network caching has been largely unexplored.

The work in [6] presents the first experimental study of NDN in a life-size IoT deployment. The authors evaluate the impact of caching by comparing performance of content retrieval from different consumers in presence of standard NDN in-network caching and when caching is disabled. Results show that, thanks to caching, the number of radio transmissions is drastically reduced. In fact, although the cache size of a resource-constrained node is at most in the order of 1KByte and information is ephemeral, caching is still beneficial: it increases the content availability and shortens the number of lossy multi-hop wireless paths towards the producer. Moreover, if, on one hand, the cache size of IoT devices is small, on the other hand, also the IoT contents' size is usually small and therefore a single cache can store several IoT packets.

In [7], the impact of IoT information freshness over NDN caching is analysed. In addition to the freshness parameter included in Data packets, which establishes how many seconds the content can be considered as valid in the CS, the authors propose a consumer driven freshness approach. This mechanism allows the consumer to specify its particular requirement in term of freshness in the Interest thus the CS can determine whether or not it has the information available. As a result, the authors prove that the accuracy of Data received by consumers is improved.

A first study on caching IoT contents in Internet content routers is [8]. It proposes a distributed probabilistic caching algorithm, where routers dynamically update their caching probability by considering their hop distances to the source and the consumers, the data freshness, the rate and freshness requirement of incoming requests. Two key concepts are the rationale of the proposal. First, IoT data become less fresh when the caching location is further away from the producer, i.e., the closer the caching location is to the source, the fresher the retrieved Data packet is. Second, retrieving Data from locations closer to the source rather than the consumer incurs higher multihop traffic load on the network. Therefore, the algorithm defines an utility function that addresses a trade-off between (i) the multihop delivery from consumer and producer and (ii) the expected loss in data freshness for the delivered Data.

Unlike [8], which focuses on the wired Internet and electrically powered, static devices, we consider an IoT multi-hop wireless network composed of (mobile) resource-constrained nodes and deploy a lightweight caching decision algorithm, as

discussed in the next Section.

III. THE CACHING DECISION STRATEGY

A. Assumptions

In our scenario, a number N of resource-constrained nodes with limited battery energy are connected in a multi-hop wireless network. Nodes can be fixed, e.g., monitoring sensors deployed in streets, bridges, buildings, or mobile, e.g., smartphones carried by humans. We assume that a set of nodes $C = \{1, \dots, N_c\}$, with $N_c < N$ act as consumers, while a producer P generates IoT contents with a specific freshness value. The rest of the nodes can act as forwarders which are candidate to cache incoming Data packets.

As discussed in [7], a consumer asks for a Data packet by broadcasting an Interest that also specifies the freshness requirement, which is fundamental in presence of IoT transient contents. Any node (different from P) that receives the request and does not have the proper Data in its CS can re-broadcast the packet according to the implemented routing protocol.

When a node receives a Data packet with a PIT matching, it performs the caching decision by leveraging the pCASTING strategy.

As fundamental requirements of the proposed solution we target *simplicity* and *no overhead*. pCASTING is simple enough to be implemented in resource-constrained devices and it does not require additional signalling or further information piggybacked in Interest/Data packets. Moreover, pCASTING is totally independent on the selected routing protocol: it can work in presence of flooding-based, producer-aware or path-aware approaches [15].

The caching decision is totally distributed and consists of two main steps that are locally computed by each forwarding node:

1. *the computation of the caching probability*, which takes into account dynamic IoT attributes related to device and content. The selected attributes are properly normalized to make heterogeneous values comparable with each other.
2. *the actual probabilistic decision*, which specifies if caching or not the Data on the basis of the computed probability.

B. Attributes description

As device attributes, we consider two main parameters: *energy level* and *cache occupancy*, whose values can be easily monitored by IoT devices.

Energy consumption is one of the main limiting factors of IoT devices. If the residual battery life falls below a pre-determined level, the activity of the device should be strongly restricted. This means that caching (and consequent transmission) operations should be inhibited as much as possible. Therefore, we assume that the caching probability is *directly proportional* to the *residual battery energy level*, which can be dynamically represented as a normalized parameter EN , with $0 \leq EN \leq 1$, where the values 0 and 1 mean that the battery is fully discharged or charged, respectively.

Concerning the cache occupancy, it is reasonable to assume that nodes with high storage capacity are more prone to cache incoming Data, while nodes with limited storage should be strongly selective in the caching decision. Intuitively, the caching probability in a node should be *inversely proportional* to the *cache occupancy level*, which can be represented as a normalized parameter OC , with $0 \leq OC \leq 1$, where the values 0 and 1 mean that the cache is empty or full, respectively.

As content attribute, we consider the Data *residual freshness*. According to NDN, any producer may include in the Data packet a value indicating the freshness f in seconds, and a timestamp t_s which identifies the instant when the information is produced.

Although nodes automatically discard from their CS every expired Data, *fresher* and *less fresh* versions of the same Data could co-exist in the network [8] and consumer applications can be interested in retrieving the same Data with different freshness requirements. For instance, a consumer C_1 could require the value of pollution in a road segment with a freshness of at most 30 minutes and, at the same time, another consumer C_2 could require the same data but with a freshness of at most 10 minutes. A node receiving both Interests could accommodate the first request with a cached copy of the Data, but forward the second request because the owned Data is not fresh enough for C_2 . This will also imply that, for a certain period, two version of the same Data with different freshness values will be available in the network.

Fresher data packets should have priority in the caching decision, since they can more likely accommodate the consumer's freshness requirement. Therefore, we define the normalized residual freshness FR of a Data packet as:

$$FR = 1 - \frac{\text{currentTime} - t_s}{f} \quad (1)$$

According to Eq. 1, if the timestamp t_s is equal to the current time then $F = 1$, and the Data can be considered as freshly generated, while, when the current time increases, the residual freshness decreases. Of course, a packet with a negative FR is expired and it is not cached at all.

C. The caching probability

In order to properly define the caching probability of a Data packet, we consider a *Caching Utility Function* F_u that simultaneously takes into account the above normalized parameters, EN , OC , FR . The Function can be written as follows:

$$F_u = \sum_{i=1}^{N_p} w_i g(x_i) \quad (2)$$

where $N_p = 3^1$ and the weights w_i assume a value such that $0 \leq w_i \leq 1$ and $\sum_{i=1}^N w_i = 1$. Therefore, the weights

¹The definition of F_u is quite flexible: it can be extended to take into account other parameters for the caching decision in addition to EN , OC , FR .

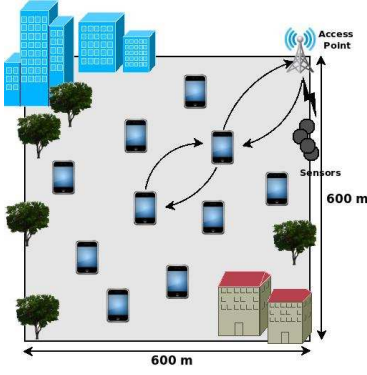


Fig. 1. Simulation scenario.

express the importance that the normalized parameter x_i has in the computing of the utility value.

F_u must assume values in the interval $[0 : 1]$ and give as a result the node's caching probability: if $F_u \rightarrow 1$ the probability that a packet is cached is high, vice versa if $F_u \rightarrow 0$ the same probability is low.

The shape of function $g(x_i)$ can be devised by considering the requirements that it should satisfy. First, it should furnish a value in-between zero and one and has to be increasing (decreasing) monotone when x_i value is directly (inversely) proportional to the overall utility of the caching. Second, it should also meet the condition that the closer the parameter x_i is to the limit values, the higher is the impact on the resulting utility. By doing so, the caching decision favours fresher contents, which can be more likely stored in nodes with the higher energy and the major cache capability.

The mathematical function satisfying such properties is the power function $g(x_i) = x_i^n$ with $n \geq 1$ chosen according to the relevance associated to the second requirement. The Utility Function is therefore expressed by the weighted sum of power functions:

$$F_u = w_1 * EN^n + w_2 * (1 - OC)^n + w_3 * FR^n \quad (3)$$

At the reception of a Data packet, any node computes the value of F_u to obtain the caching probability of that Data.

IV. EXPERIMENTS

A. Simulation setup

To evaluate the performance of the proposed pCASTING solution we consider a smart city scenario in Fig. 1 representing an urban area of size 600m x 600m.

A group of sensors deployed in the city periodically generates environmental data and other location-dependent information. Collection points retrieve those contents and manipulate them to create context-aware services for users, e.g., real-time public transportation updates, air pollution updates. An Access Point (AP), placed at the upper right corner of the square topology, offers to interested consumers such context-aware services. We assume that 60 nodes move in the scenario at a pedestrian speed according to the Truncated Levy-Walk mobility model [16], which generates realistic human mobility

TABLE I
SIMULATION SETTINGS.

Category	Parameter	Value
Application	Data packet size	512 bytes
	Consumer's freshness	rand(1:10)
	Consumer's update period	1 minute
	N_d	5
NDN	CS size	10 packets
	Routing	controlled flooding
	pCASTING	$w_1 = w_2 = w_3 = 1/3$
		$n = 1$
Access	Technology	IEEE 802.11g
	Rx Sensivity	-83 dBm
	Propagation	Rayleigh
Scenario	Area size	600m x 600m
	Number of mobile nodes	60
	Mobility Model	pedestrian [16]
	Number of consumers	1-8

traces. Such mobile nodes mimic the smartphones carried by users that want to access the AP's services. Each node is equipped with the IEEE 802.11g radio interface where medium access control and physical parameters (e.g., transmission power and receiver sensitivity) are set according to datasheets of devices today on the market [17]. Propagation is modelled through the Rayleigh fading distribution to account for multipath effects due to various obstacles, buildings and trees.

A subset of nodes in the topology is selected for subscribing to the context-aware services offered by the AP. Each service is modelled as a group of N_d Data packets, i.e., each consumer must transmit N_d different Interests to obtain the overall information. However, since information is transient, each consumer must periodically send novel Interests to obtain the fresher information. In our scenario, the update period is 1 minute (e.g., the consumer requests the content at $t = 0s$, $t = 60s$, $t = 120s$, etc.) and each time the Interests also carry the freshness requirement, which is randomly chosen in the range of $[1 : 10]$ seconds. Communication with the AP is performed in a multi-hop fashion. Without loss of generality, the forwarding protocol we refer to is the controlled flooding, as defined in [15], [6] specifically for wireless networks. It consists of a counter-based broadcasting scheme which defers packet forwarding in order to limit collision probability and redundancy on the wireless shared medium. Controlled-flooding can be a good choice in IoT networks due to its intrinsic simplicity: it does not rely on additional control traffic and requires minimal state at nodes. Moreover, in a medium/small topology (as the one considered in this paper) the potential overhead due to redundant transmissions will be low.

To study the behaviour of the caching strategy in presence of high storage constraint, we assume that the CS capacity in each node is limited to only 10 packets.

The described scenario and the pCASTING solution were implemented in the ndnSIM tool [10], the reference simulation platform of the NDN research community deployed in ns-3 [18]. Simulation settings are summarized in Table I. In the following, we consider two different experiments that aim to assess the energy efficiency of the proposed solution and the

data retrieval performance.

B. Network lifetime analysis

The first target of our evaluation is to understand if pCASTING is an energy-efficient solution in presence of battery-powered devices by also guaranteeing good performance in terms of retrieval delay and number of collected Data. Therefore, the *energy model* in ns-3 has been used to calculate the energy consumption of wireless devices in the simulated scenario. The model allows to specify the *initial energy* of each device and during the simulation it decreases such value on the basis of the activity of the node.

In the experiment, eight nodes are selected as consumers and provided with a high level of initial energy, while forwarding nodes are provided with a very low level of initial energy and they are meant to quickly run out of energy. Our main focus, in fact, is to estimate the overall *network lifetime* by considering the energy consumption of the forwarding nodes, which allow the communication between consumers and the AP. This latter offers four different context-aware services: each consumer randomly selects one of the services² and starts requesting Data at some random instants chosen uniformly within the intervals $[0 : 5]$ seconds. Then, at every minute, each consumer sends again the Interests. Simulation ends when all the forwarding nodes have exhausted their energy.

We compare pCASTING against three reference schemes: (i) the Caching Everything Everywhere strategy, CE^2 , which caches every incoming packet; (ii) a probabilistic scheme, $P(0.5)$, which caches incoming Data with fixed probability $p = 0.5$; (iii) the *no caching* scheme, where nodes do not use their CS but only forward incoming packets. The replacement policy is LRU for every caching scheme.

As energy performance metric, we consider the *cumulative distribution function* (CDF) of the *discharge time of the forwarding nodes*, which gives information about the energy efficiency of the caching strategy from the network point of view. As indicators of the dissemination performance, we consider the *cache hit ratio*, the *average number of Data packets received by the consumers* during the network lifetime and the *average data retrieval delay*, calculated as the time interval when the Interest is sent and the correspondent Data is received. The delay takes into account possible Interest retransmissions.

Results are calculated over ten independent runs.

Fig. 2 shows the CDF of the discharge time of the forwarding nodes, while Table II shows the values of the considered data dissemination performance metrics. It can be observed that the shortest network lifetime is obtained with the CE^2 scheme: at instant $t = 320s$ the CDF reaches the value 1, which means that all the forwarding nodes run out of energy. This is due to the fact that CE^2 generates a lot of caching/forwarding operations in the nodes and, at the same time, it does not maximize the content diversity. In the wireless

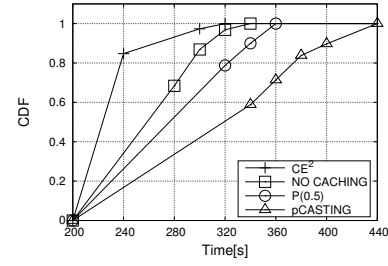


Fig. 2. CDF of the discharge time of the forwarding nodes.

TABLE II
DATA DISSEMINATION PERFORMANCE METRICS BY VARYING THE CACHING STRATEGY.

Metric	CE^2	No Caching	$P(0.5)$	pCASTING
Cache hit ratio	42%	0	49%	61%
Consumers' received data pkts	206	208	217	239
Data retrieval delay[s]	0.2	0.34	0.18	0.12

domain, in fact, many neighbouring nodes are supposed to receive the same Data packet at the same time instant and, as a consequence of the CE^2 scheme, they all cache the same information. This condition is further exacerbated in presence of small CS, which causes a high eviction rate and creates a highly homogeneous content distribution over time. The presence of such homogeneous cached contents has a twofold side effect. On one hand, requests for different contents cannot be satisfied by any intermediate node, and Interests must be forwarded to the original producer (see the CE^2 cache hit ratio in Table II). On the other hand, in presence of a CS matching, many nodes at the same time can answer to the same request by uselessly wasting energy and network resources. It also worth noticing that Data packets are usually longer than Interests and therefore the probability of Data losses for channel errors or collisions is higher than Interest losses.

The *no caching* scheme can offer a slightly higher network lifetime compared to CE^2 , but reasonably it experiences the higher end-to-end delay, since the benefits of caching cannot be exploited in any way.

Better performance is obtained with the $P(0.5)$ scheme which, by introducing the probabilistic caching decision, increases the content diversity in the network and facilitates cache hits at intermediate nodes. However, thanks to its energy-aware behaviour, pCASTING guarantees the highest network lifetime and, as a further consequence, the highest number of retrieved data packets. Moreover, thanks to the intrinsic probabilistic decision, pCASTING maximizes the content diversity in the network thus increasing the cache hit ratio and, consequently, reducing the data retrieval delay.

C. Retrieval performance analysis

The second experiment studies the performance of the data dissemination in presence of fully charged nodes. The ns-3 energy model is still applied to show the energy consumption

²It is worth noticing that we do not refer to the Zipf distribution for modelling the access to contents since this distribution is devised for web-based contents and Internet applications, which are not considered here.

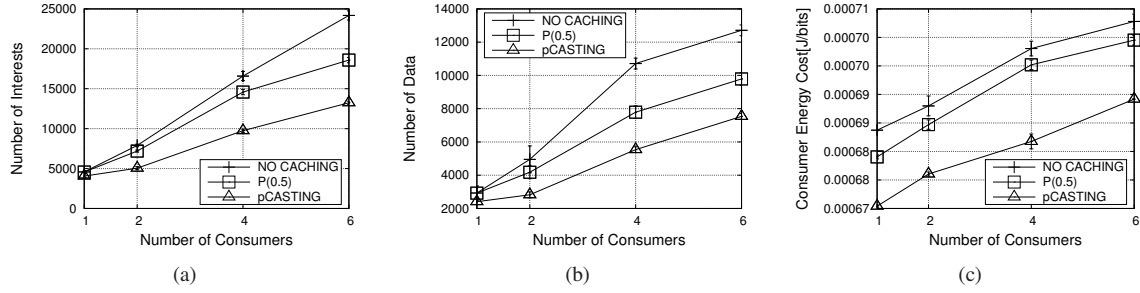


Fig. 3. Metrics for retrieval performance analysis: (a) Number of Interest transmitted in the network; (b) Number of Data transmitted in the network; (c) Consumer Energy Cost.

during the simulation, which lasts 10 minutes. In such period, a variable number of consumers ranging from 1 to 6 periodically asks for context-aware services. Simulation results are averaged over 10 independent runs and reported with the 95% confidence intervals.

We compare pCASTING against the *no caching* and *P(0.5)* schemes. To evaluate the efficiency of the caching strategy from the network point of view, we consider the *number of Interests* and the *number of Data packets* transmitted in the network during the simulation time. Vice versa, to evaluate the efficiency of the caching strategy from the consumer point of view, we calculate the *consumer energy cost*, which is the ratio between the energy spent by a consumer and the correctly received bits, averaged on all the consumers.

A glance at the number of Interest and Data packets in Fig. 3(a) and 3(b), respectively, indicates that pCASTING strongly reduces the amount of traffic compared to the other schemes, thus saving network resources. But also the consumers have advantages from pCASTING, since it reduces the energy consumption, as shown in Fig. 3(c). This is due to the fact that, being aware of the status of the node (in term of energy and cache size) and the residual freshness of the Data packets, pCASTING performs a smart caching decision which efficiently distributes the contents in the network. As a further result, the pCASTING cache hit ratio³ is higher than the agnostic probabilistic scheme *P(0.5)* by maintaining in the order of 55-62%, while the cache hit ratio of *P(0.5)* is always under 50%.

V. CONCLUSION

In this paper, we studied in-network caching in named-data wireless IoT networks and proposed a distributed caching scheme called pCASTING that adjusts the caching probability by considering three main parameters: the battery energy level and the cache occupancy of the node, and the Data packet's residual freshness. Simulation results proved the effectiveness and efficiency of the proposed solution, which is able to reduce the energy consumption in the nodes and, at the same time, guarantee low content retrieval delays.

The way we compute the caching probability in pCASTING is very versatile: the Caching Utility Function F_u can work

with different routing protocols and it can be easily modified by adding other parameters. Indeed, as a future work, we plan (i) to include other attributes in the caching decision, and (ii) to evaluate the possible interactions between caching and different routing protocols.

VI. ACKNOWLEDGEMENT

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³We do not present the related plot here due to space limitations.